Dear Editor,

Thank you very much for arranging to review our manuscript. We have carefully addressed all the questions and suggestions raised by Reviewer #1. A detailed, point-by-point response is provided below (the comments of the reviewer are highlighted in blue). In view of these changes, we trust the manuscript will now be judged suitable for publication.

Sincerely,
Author

## I. RESPONSE TO QUESTIONS/COMMENTS OF REVIEWER #1

1. This manuscript systematically investigates the linear magnetoelectric conductivity and nonlinear electrochemical response (ECR) of three classes of three-dimensional topological semimetals, namely vortex nodal-ring semimetals (VNRs), $\mathcal{PT}$-symmetric nodal-ring semimetals (PTNRs), and three-band Hopf semimetals (HSMs), within the semiclassical Boltzmann transport formalism. Most existing topological-transport literature concentrates on Berry-curvature monopoles in Weyl nodes. Transport phenomena driven by higher-order multipole sources such as Berry-curvature dipoles remain comparatively underexplored. This work unifies VNRs (continuously distributed dipoles along the nodal ring) and HSMs (isolated point-like dipoles) within a common theoretical framework, compares their transport properties side-by-side with PTNRs, and advances our understanding of topological transport in zero-Chern-number systems. The topic is theoretically relevant, physical reasoning is sound, and derivations are largely complete. Nevertheless, several points require revision before acceptance.

   We thank the reviewer for the positive assessment of the topic, of the unified treatment of VNRs and HSMs, and of the physical reasoning, and for the constructive points raised below. All the resulting additions are marked in red in the revised manuscript.

2. The manuscript states that VNRs and HSMs lack conventional time-reversal symmetry and obey a composite antiunitary symmetry $\Xi = T\,M_y$ instead, which lifts the Onsager-Casimir constraint forbidding odd-in-$B_z$ terms. A concise explanatory paragraph should be added to clarify the distinction between "time-reversal symmetry of the bare Hamiltonian" and the "generalised parameter-reversal relation for transport coefficients".

   We agree that the two notions were discussed in separate places and that their logical relation deserved a dedicated paragraph. We have added the following text at the end of the symmetry discussion of the VNR in Sec. II, after the discussion of Eq. (19):
   "The time-reversal symmetry of the bare Hamiltonian and the parameter-reversal relation obeyed by the transport coefficients are logically distinct statements. The former is a property of a fixed Hamiltonian, namely the existence of an antiunitary operator $T$ with $T\,\mathcal{H}(\boldsymbol{k})\,T^{-1} = \mathcal{H}(-\boldsymbol{k})$. The latter follows from the microscopic reversibility of the full dynamics and relates the conductivity of a system to that of its time-reversed partner, in which every time-reversal-odd parameter is reversed. In the Boltzmann response, these parameters are $\boldsymbol{B}$, the tilt $\boldsymbol{\eta}$, and the geometric fields $\boldsymbol{\Omega}_s$ and $\boldsymbol{m}_s$, which gives Eq. (18). When $\mathcal{H}$ is itself time-reversal invariant, as for the PTNR, the reversal of $\boldsymbol{\Omega}_s$ and $\boldsymbol{m}_s$ is equivalent to the original system, and Eq. (18) closes on a single system. For the diagonal component it then reduces to the familiar Onsager-Casimir form $\sigma_{zz}(B_z) = \sigma_{zz}(-B_z)$, which forbids odd powers of $B_z$. In a single VNR or a single HSM node, the reversal of $\boldsymbol{\Omega}_s$ and $\boldsymbol{m}_s$ flips the orientation of the Berry-curvature dipole and therefore produces a different system. Eq. (18) then relates $\sigma_{zz}$ of the system to that of its dipole-reversed partner and not to itself, so that no constraint on the parity of $\sigma_{zz}$ in $B_z$ results. The surviving antiunitary symmetry $\Xi$ yields only the covariance relation of Eq. (19), which reduces to an identity for $\boldsymbol{E} \parallel \boldsymbol{B} \parallel \hat{\boldsymbol{z}}$. Odd powers of $B_z$ are thus symmetry-allowed, and whether they actually appear is decided by the parity of the integrand under $k_z \to -k_z$, as discussed in Sec. III."

3. What behavior is expected for the odd-in-$B_z$ terms of linear magnetoconductivity in the limit $\mu \to 0$, i.e., when the chemical potential approaches the node/nodal ring? Are there any emergent singularities in this limit? This limit is not explicitly discussed in the current manuscript.

   We thank the reviewer for raising this point, which we had not addressed. The behaviour is controlled by a single dimensionless parameter for each system, as we now explain in a new subsection at the end of Sec. III. The text reads: "Let us comment on the limit $\mu \to 0$ at fixed $B_z$. For the HSM node, the Zeeman-like shift gives $\xi_2(\boldsymbol{k}) = v_0\,k + e\,v_0\,B_z\cos^2\gamma/(2\,k)$. The only scales are $\mu$ and $v_0\sqrt{e\,B_z}$, so that $\delta\sigma_{zz}$ is a function of the single parameter

$$\zeta_{\rm h} = \frac{e\,B_z\,v_0^2}{\mu^2}\,. \tag{1}$$

   The coefficient of the $n^{\rm th}$ power of $B_z$, whether odd or even, therefore scales as $\mu^{-2n}$ at fixed $B_z$. The weak-field expansion is thus controlled only for $\zeta_{\rm h} \ll 1$. For $B_z > 0$, the minimum of $\xi_2$ along the direction $\gamma$ equals $v_0\sqrt{2\,e\,B_z}\,|\cos\gamma|$, which coincides with the reality condition of $k_F$ in Sec. III B and corresponds to $\zeta_{\rm h} = 1/2$. Below the threshold $\mu_c = v_0\sqrt{2\,e\,B_z}$, the Fermi surface opens at the polar caps and the sphere turns into a surface of toroidal topology. We

therefore expect the exact $\delta\sigma_{zz}$ to be nonanalytic at $\zeta_{\rm h} = 1/2$ for $B_z > 0$, with a square-root branch point inherited from $k_F$. No such threshold exists for $B_z < 0$, where the OMM lowers the energy. For the VNR, the Zeeman-like shift reduces at leading order in $\kappa/k_0$ to $e\,v_0\,B_z\sin^2\gamma/(2\,k_0)$, which does not depend on $\kappa$. The Fermi momentum becomes $\kappa_F = \mu/v_0 - e\,B_z\sin^2\gamma/(2\,k_0)$, and both the OMM and the BC corrections organise in powers of

$$\zeta_{\rm v} = \frac{e\,v_0\,B_z}{2\,k_0\,\mu}\,. \tag{2}$$

The expansion coefficients then grow as $\mu^{-n}$, which is milder than for the HSM. For $B_z > 0$, the tube of the toroidal Fermi surface pinches off at $\gamma = \pm\pi/2$ when $\zeta_{\rm v} = 1$. The divergence of the BC at the ring and at the node is integrable and does not generate a singularity of its own. The growth of the coefficients, together with the pinch-off, signals the breakdown of the weak-field expansion and not a physical divergence of the response. Moreover, a semiclassical treatment requires $\mu \gg v_0\sqrt{e\,B_z}$, so that the regime $\zeta \gtrsim 1$ for the HSM lies in the quantum limit, where a Landau-level description is needed. For the VNR, $\zeta_{\rm v} \ll 1$ remains satisfied down to this scale for weak fields. Finite temperature additionally cuts off the growth once $\mu \lesssim T$. For the PTNR, the BC and OMM are supported only on the ring, and no odd-in-$B_z$ term arises on account of time-reversal symmetry."

4. If both in-plane and axial tilts are present simultaneously for ECR responses, do interference effects emerge among different non-vanishing tensor components?

We thank the reviewer for this question. We have carried out the calculation of the tensors $t^s_{ab}$ for $\boldsymbol{\eta} = (\eta_x, 0, \eta_z)$ explicitly for all three systems, and we have replaced the earlier qualitative discussion by explicit analytical results. The intermediate steps are given in the new Appendix B of the manuscript. The subsection at the end of Sec. IV now reads as follows:
"Let us finally consider $\boldsymbol{\eta} = \eta_x\,\hat{\boldsymbol{x}} + \eta_z\,\hat{\boldsymbol{z}}$. We define $\eta \equiv |\boldsymbol{\eta}| = \sqrt{\eta_x^2 + \eta_z^2}$ and $\hat{\eta}_{x,z} \equiv \eta_{x,z}/\eta$, and we restrict ourselves to $\mu > 0$ and $\eta < 1$. Eq. (71) remains valid for an arbitrary tilt, and only the evaluation of $t^s_{ab}$ is required. The intermediate steps of the calculation are collected in the new Appendix B.

This tilt breaks both the $C_{\infty z}$ rotation and the $k_z$ mirror. The antiunitary symmetry $\Xi$ survives since the tilt has no $y$ component. For the VNR and the HSM, the components $\Omega^{x,z}_s$ and $v_{x,z}$ are even under $k_y \to -k_y$, whereas $\Omega^y_s$ and $v_y$ are odd, and $f_0'$ is even. Hence $t^s_{xy}$, $t^s_{yx}$, $t^s_{yz}$ and $t^s_{zy}$ vanish, while $t^s_{xx}$, $t^s_{yy}$, $t^s_{zz}$, $t^s_{xz}$ and $t^s_{zx}$ are allowed. The mirror and the rotation by $\pi$ about $\hat{\boldsymbol{z}}$ further show that the diagonal components are even in $\eta_x$ and odd in $\eta_z$, whereas $t^s_{xz}$ and $t^s_{zx}$ are odd in $\eta_x$ and even in $\eta_z$. We now present the explicit forms of these five components, starting with the HSM.

For the HSM, only the $s = 2$ band has a Fermi surface at $\mu > 0$. Defining the three functions

$$\begin{aligned}
\mathcal{F}_1(\eta) &= \frac{3\,(1-\eta^2)\,\operatorname{arctanh}\eta - 3\,\eta + 2\,\eta^3}{3\,\eta^4}\,, \quad \mathcal{F}_2(\eta) = \frac{3\,(1-\eta^2)^2\,\operatorname{arctanh}\eta - 3\,\eta + 5\,\eta^3}{3\,\eta^4}\,,\\
\mathcal{F}_3(\eta) &= \frac{3\,(1-\eta^2)\,(\eta^2-5)\,\operatorname{arctanh}\eta + 15\,\eta - 13\,\eta^3}{3\,\eta^4}\,,
\end{aligned} \tag{3}$$

each of which vanishes linearly as $\eta \to 0$, we find

$$\begin{aligned}
t^2_{xx} &= -\frac{\hat{\eta}_z}{4\,\pi^2}\left[\mathcal{F}_1 + \mathcal{F}_3\,\hat{\eta}_x^2\right], \quad t^2_{yy} = -\frac{\hat{\eta}_z}{4\,\pi^2}\,\mathcal{F}_1\,, \quad t^2_{zz} = -\frac{\hat{\eta}_z}{4\,\pi^2}\left[2\,\mathcal{F}_1 + \mathcal{F}_2 + \mathcal{F}_3\,\hat{\eta}_z^2\right],\\
t^2_{xz} &= -\frac{\hat{\eta}_x}{4\,\pi^2}\left[\mathcal{F}_2 + \mathcal{F}_3\,\hat{\eta}_z^2\right], \quad t^2_{zx} = -\frac{\hat{\eta}_x}{4\,\pi^2}\left[\mathcal{F}_1 + \mathcal{F}_3\,\hat{\eta}_z^2\right].
\end{aligned} \tag{4}$$

For $\eta_x = 0$ these reduce to the results of Case II, and for $\eta_z = 0$ to those of Case I, including the relation $D^2_{zx} = -\,t^2_{zx}$. The results are independent of $\mu$, as for the single tilts. The three exact relations

$$t^2_{xx} + t^2_{yy} + t^2_{zz} = 0\,, \quad t^2_{xx} - t^2_{yy} = -\,\frac{\hat{\eta}_x^2\,\hat{\eta}_z}{4\,\pi^2}\,\mathcal{F}_3\,, \quad t^2_{xz} - t^2_{zx} = -\,\frac{\hat{\eta}_x}{4\,\pi^2\,\eta^2}\left[\eta - (1-\eta^2)\,\operatorname{arctanh}\eta\right] \tag{5}$$

follow directly. The vanishing of the trace holds for any tilt, since $(\boldsymbol{n} + \boldsymbol{\eta})\cdot\boldsymbol{n} = 1 + \boldsymbol{\eta}\cdot\boldsymbol{n}$, with $\boldsymbol{n} = \hat{\boldsymbol{k}}$, cancels the denominator of the integrand and leaves an integral of $n_z$ over the sphere. Hence, as soon as both tilts are present, the isotropy $t^2_{xx} = t^2_{yy}$ is lifted, and the relation $t^2_{zz} = -2\,t^2_{xx}$ of Case II is replaced by $t^2_{zz} + 2\,t^2_{xx} = t^2_{xx} - t^2_{yy}$. At small tilt, the expansions

$$\begin{aligned}
t^2_{xx} &= \frac{1}{\pi^2}\left[\frac{\eta_z}{30} + \frac{\eta_z^3}{70} - \frac{\eta_x^2\,\eta_z}{42}\right] + \mathcal{O}(\eta^5)\,, \quad t^2_{yy} = \frac{1}{\pi^2}\left[\frac{\eta_z}{30} + \frac{\eta_z^3}{70} + \frac{\eta_x^2\,\eta_z}{70}\right] + \mathcal{O}(\eta^5)\,,\\
t^2_{zz} &= \frac{1}{\pi^2}\left[-\frac{\eta_z}{15} - \frac{\eta_z^3}{35} + \frac{\eta_x^2\,\eta_z}{105}\right] + \mathcal{O}(\eta^5)\,,\\
t^2_{xz} &= \frac{1}{\pi^2}\left[-\frac{2\,\eta_x}{15} - \frac{2\,\eta_x^3}{105} - \frac{2\,\eta_x\,\eta_z^2}{35}\right] + \mathcal{O}(\eta^5)\,, \quad t^2_{zx} = \frac{1}{\pi^2}\left[\frac{\eta_x}{30} + \frac{\eta_x^3}{70} - \frac{\eta_x\,\eta_z^2}{42}\right] + \mathcal{O}(\eta^5)
\end{aligned} \tag{6}$$

show that the response is additive to linear order in the tilt, with $\eta_x$ feeding only $t^2_{xz}$ and $t^2_{zx}$, and $\eta_z$ only the diagonal components. The interference between the two tilt directions is captured by $\Delta_{ab} \equiv t^2_{ab}(\eta_x,\eta_z) - t^2_{ab}(\eta_x,0) - t^2_{ab}(0,\eta_z)$. It starts at third order, with

$$\Delta_{xx} = -\frac{\eta_x^2\,\eta_z}{42\,\pi^2}\,, \quad \Delta_{yy} = \frac{\eta_x^2\,\eta_z}{70\,\pi^2}\,, \quad \Delta_{zz} = \frac{\eta_x^2\,\eta_z}{105\,\pi^2}\,, \quad \Delta_{xz} = -\frac{2\,\eta_x\,\eta_z^2}{35\,\pi^2}\,, \quad \Delta_{zx} = -\frac{\eta_x\,\eta_z^2}{42\,\pi^2}\,, \tag{7}$$

up to corrections of order $\eta^5$. No component is proportional to the product $\eta_x\,\eta_z$, and the interference is therefore of the type $\eta_x^2\,\eta_z$ for the diagonal components and $\eta_x\,\eta_z^2$ for the off-diagonal ones.

For the VNR, only the $s = 2$ band contributes if $\mu \geq \tilde{\eta}$, where $\tilde{\eta} = v_0\,k_0\,\eta_x$. In that regime, the results again do not depend on $\mu$ and are given by

$$t^2_{ab} = \frac{1}{16\,\pi^3}\int_0^{2\,\pi} d\phi\;\mathscr{S}_{ab}(\phi)\,, \tag{8}$$

where the integrands $\mathscr{S}_{ab}(\phi)$ are elementary functions of $R(\phi) = \sqrt{\eta_z^2 + \eta_x^2\cos^2\phi}$ and $\sqrt{1-R^2(\phi)}$, listed in Eqs. (B10) and (B14) of the new Appendix B. The condition $\eta < 1$ is precisely the requirement that $1 - R^2(\phi) > 0$ for all $\phi$, which ensures that the toroidal Fermi surface remains closed.

For $\mu \geq \tilde{\eta}$, the remaining $\phi$-integral can be performed in closed form. With $\varrho = \sqrt{1-\eta_z^2}$, $k^2 = \eta_x^2/\varrho^2$ and $\hat{\Pi} \equiv \Pi\big(-\eta_z^2/\varrho^2\,\big|\,k^2\big)$, where $K$, $E$ and $\Pi$ denote the complete elliptic integrals of the first, second and third kind, we obtain

$$t^2_{ab} = \frac{1}{4\,\pi^2}\left[\mathcal{A}_{ab}\,K(k^2) + \mathcal{B}_{ab}\,E(k^2) + \mathcal{C}_{ab}\,\hat{\Pi}\right] + \mathcal{E}_{ab}\,, \tag{9}$$

with the coefficients $\mathcal{A}_{ab}$, $\mathcal{B}_{ab}$, $\mathcal{C}_{ab}$ and $\mathcal{E}_{ab}$ listed in the new Appendix B. For $\eta_x \to 0$ and $\eta_z \to 0$, the closed form above reduces to the results of Case II and Case I, respectively. The single-tilt results of the VNR and the HSM are thus the limits $\eta_z \to 0$ and $\eta_x \to 0$ of Eqs. (100) and (95), where the former holds for $\mu \geq \tilde{\eta}$. The regime $\mu < \tilde{\eta}$ of the VNR is covered by Eq. (103). The trace identity holds for the same reason as in the HSM. The expansions

$$\begin{aligned}
t^2_{xx} &= \frac{1}{\pi}\left[-\frac{\eta_z}{64} - \frac{\eta_z^3}{128} + \frac{7\,\eta_x^2\,\eta_z}{512}\right] + \mathcal{O}(\eta^5)\,, \quad t^2_{yy} = \frac{1}{\pi}\left[-\frac{\eta_z}{64} - \frac{\eta_z^3}{128} - \frac{3\,\eta_x^2\,\eta_z}{512}\right] + \mathcal{O}(\eta^5)\,,\\
t^2_{zz} &= \frac{1}{\pi}\left[\frac{\eta_z}{32} + \frac{\eta_z^3}{64} - \frac{\eta_x^2\,\eta_z}{128}\right] + \mathcal{O}(\eta^5)\,,\\
t^2_{xz} &= \frac{1}{\pi}\left[\frac{7\,\eta_x}{64} + \frac{5\,\eta_x^3}{512} + \frac{9\,\eta_x\,\eta_z^2}{128}\right] + \mathcal{O}(\eta^5)\,, \quad t^2_{zx} = \frac{1}{\pi}\left[-\frac{\eta_x}{64} - \frac{3\,\eta_x^3}{512} + \frac{\eta_x\,\eta_z^2}{128}\right] + \mathcal{O}(\eta^5)
\end{aligned} \tag{10}$$

lead to the same structure as for the HSM. The response is additive to linear order, and the interference terms are

$$\Delta_{xx} = \frac{7\,\eta_x^2\,\eta_z}{512\,\pi}\,, \quad \Delta_{yy} = -\frac{3\,\eta_x^2\,\eta_z}{512\,\pi}\,, \quad \Delta_{zz} = -\frac{\eta_x^2\,\eta_z}{128\,\pi}\,, \quad \Delta_{xz} = \frac{9\,\eta_x\,\eta_z^2}{128\,\pi}\,, \quad \Delta_{zx} = \frac{\eta_x\,\eta_z^2}{128\,\pi}\,. \tag{11}$$

For $\mu < \tilde{\eta}$, the $\phi$-integration is restricted, as in the single-tilt cases. With $\phi_m \equiv \arccos\left[\min(1,\mu/\tilde{\eta})\right]$, we obtain

$$t^2_{ab} = \frac{1}{16\,\pi^3}\int_{\phi_m}^{2\,\pi-\phi_m} d\phi\;\mathscr{S}_{ab}(\phi)\,, \quad t^1_{ab} = -\frac{1}{16\,\pi^3}\int_{-\phi_m}^{\phi_m} d\phi\;\mathscr{S}_{ab}(\phi)\,, \tag{12}$$

so that the band sum is given by $t^1_{ab} + t^2_{ab} = t^2_{ab}\big|_{\mu\geq\tilde{\eta}} - \frac{1}{8\,\pi^3}\int_{-\phi_m}^{\phi_m} d\phi\;\mathscr{S}_{ab}(\phi)$. The trace vanishes for every $\mu$, since the $\phi$-restriction does not affect the identity $\int_0^{2\pi} d\gamma\,\sin\gamma = 0$. The relations $\eta_x^2 + \eta_z^2 < 1$ for the VNR and $|\boldsymbol{\eta}| < 1$ for the HSM play the same role, namely that of keeping the effective tilt in each cross-section of the Fermi surface below unity.

For the PTNR, all the contributions are supported on the ring, where the energy is $\varepsilon_s = \tilde{\eta}\cos\phi$ irrespective of $\eta_z$. As in Case I, only the tilt part $v_0\,\boldsymbol{\eta}$ of the velocity survives on the ring, and the BC is $\propto \hat{\boldsymbol{\phi}}$. Hence $t^s_{ab}$ is nonzero only for $b = y$ and $a \in \{x,z\}$, with

$$t^s_{ay} = (-1)^s\,\frac{\eta_a}{\eta_x}\,\frac{\mu}{4\,\pi^2\,\sqrt{\tilde{\eta}^2-\mu^2}}\,\Theta(\tilde{\eta}-\mu)\,, \quad a \in \{x,z\}\,. \tag{13}$$

The component $t^s_{xy}$ coincides with $-D^s_{xy}$ of Case I, and $t^s_{zy} = (\eta_z/\eta_x)\,t^s_{xy}$ is new. It exists only if $\tilde{\eta} \geq \mu$, which explains why a pure axial tilt gives no response. Since $t^1_{ay} = -\,t^2_{ay}$, the band-summed ECR of the PTNR continues to vanish for the combined tilt.

Finally, the nonzero three-index coefficients follow from Eq. (71). For the VNR and the HSM, they are

$$\begin{aligned}
&\vartheta^{(s,1)}_{xyx} = -\,\vartheta^{(s,1)}_{yxx} = e^2\,\tau\,t^s_{xz}\,, \quad \vartheta^{(s,1)}_{xyz} = -\,\vartheta^{(s,1)}_{yxz} = e^2\,\tau\,t^s_{zz}\,, \quad \vartheta^{(s,1)}_{yzx} = -\,\vartheta^{(s,1)}_{zyx} = e^2\,\tau\,t^s_{xx}\,,\\
&\vartheta^{(s,1)}_{yzz} = -\,\vartheta^{(s,1)}_{zyz} = e^2\,\tau\,t^s_{zx}\,, \quad \vartheta^{(s,1)}_{zxy} = -\,\vartheta^{(s,1)}_{xzy} = e^2\,\tau\,t^s_{yy}\,,\\
&\vartheta^{(s,2)}_{axy} = -\,\vartheta^{(s,2)}_{ayx} = -\,e^2\,\tau\,t^s_{az}\,, \quad \vartheta^{(s,2)}_{ayz} = -\,\vartheta^{(s,2)}_{azy} = -\,e^2\,\tau\,t^s_{ax}\,, \quad a \in \{x,z\}\,,\\
&\vartheta^{(s,2)}_{yzx} = -\,\vartheta^{(s,2)}_{yxz} = -\,e^2\,\tau\,t^s_{yy}\,.
\end{aligned} \tag{14}$$

The sets of nonzero components of Eqs. (79), (80) and (82) are thus active simultaneously. For the PTNR, the individual-band coefficients are $\vartheta^{(s,1)}_{zxc} = -\vartheta^{(s,1)}_{xzc} = e^2\,\tau\,t^s_{cy}$ and $\vartheta^{(s,2)}_{axz} = -\vartheta^{(s,2)}_{azx} = e^2\,\tau\,t^s_{ay}$ with $a, c \in \{x, z\}$, which cancel in the sum over $s$."

5. Captions for Figure 1, Figure 2 and Figure 3 could be expanded to visually emphasis the conceptual connection between continuously-distributed Berry-curvature dipoles in VNRs and isolated dipoles in HSMs.

We agree that this connection should be visible directly from the figures. In keeping with the journal style, we have kept the captions short and have moved the discussion to the main text. The captions now read as follows.
"Fig. 1: (a) Dispersion in the $k_xk_y$-plane, together with the pseudospin and BC profiles, of a VNR, where the BC forms a dipole in every cross-section of the nodal ring (green). (b) The corresponding plots for a Weyl node (green dot), whose BC is a monopole."
"Fig. 2: BC-flux distributions on the $k_yk_z$-plane at $k_x = 0$ for (a) a monopole, (b) an ideal dipole (HSM) and (c) a VNR. White arrows mark the dipole-like singularities, which are isolated for the HSM and continuously distributed along the ring for the VNR."
"Fig. 3: Fermi surfaces of the tilted nodal ring, which are identical for the VNR and the PTNR since the two share the same dispersion."
In addition, we have added the following paragraph to the Introduction, after the discussion of Fig. 1:
"The connection between the two dipolar systems is made explicit in Fig. 2. In the plane spanned by $\kappa_\perp$ and $k_z$ at fixed $\phi$, the BC of the VNR has the angular structure $k_z\,\boldsymbol{k}$ of the HSM, with the radial dependence of the HSM replaced by the one set by the distance from the ring. The VNR can therefore be viewed as a one-parameter family, labelled by $\phi$, of such dipoles. Since the dispersions of the VNR and the PTNR coincide (Fig. 3), the similarity of the responses of the VNR and the HSM, as opposed to the PTNR, can be traced to the BC texture and not to the shape of the Fermi surface."
We have also added the missing panel labels in Fig. 2, which are referred to in the text as Figs. 2(b) and 2(c).

## II. ADDITIONAL CORRECTIONS TO THE MANUSCRIPT

While preparing the revision, we also identified and corrected two statements in the two-node subsection of Sec. III, which concern the parity of the net response in $B_z$ and the role of the electron-number constraint. We list the corrected text below.
"For the chosen parameters, the curves of the individual nodes contain linear-in-$B_z$ terms with opposite slopes, and the linear parts dominate. The two curves are mirror images of each other, with the $\chi = +1$ curve at $B_z$ coinciding with the $\chi = -1$ curve at $-B_z$. This follows since $\xi_{\chi,2}$ depends on $\chi$ and $B_z$ only through the product $\chi\,B_z$ and the overlap function is independent of $\chi$. The sum over the two nodes is consequently even in $B_z$, so that the odd-in-$B_z$ terms cancel in the net response. The net response is therefore much smaller than the contribution of each node for these parameters. The ratio $\beta_{\rm inter}/\beta_{\rm intr}$ changes the linear terms of each node only slightly."
The earlier text had described the net curves as asymmetric about $B_z = 0$, which is correct only for the individual nodes. The corresponding statements in the Summary and the Introduction have been adjusted, and the odd-in-$B_z$ terms are now attributed to a single ring or node, or to an HSM with imbalanced nodes.
We have also replaced the earlier statement that the electron-number constraint links the two nodes and reshapes the single-node curves, which was incorrect. The revised text reads:
"As for the single node, $h_{\chi,2}$ is odd under $\gamma \to \pi - \gamma$ and contains only the odd harmonics $\cos\gamma$ and $\cos 3\gamma$. The $\chi$-dependence enters only through the product $\chi\,B_z$ in quantities that are even under $\boldsymbol{k} \to -\boldsymbol{k}$, and it does not affect this parity. Its Fermi-surface average therefore vanishes to all orders in $B$. This is the analogue of the vanishing of the charge-pumping vector $\boldsymbol{\mathcal{I}}^\chi$ in the relaxation-time approximation [**?** ]. The overlap function of Eq. (56) preserves this parity, since only the term $4\cos\gamma\cos\gamma'$ is odd, and it does not mix with the even terms. The even sector $\{\lambda_\chi, b_\chi\}$ is thus never sourced and vanishes, whereas $a_\chi$ is sourced by the $\cos\gamma$ component of $h_{\chi,2}$. We have confirmed this by solving the six-by-six system of Eq. (59), which gives $\lambda_\pm = b_\pm = 0$ for all values of $B_z$, $\mu$ and $\beta^{\rm inter}_{2,2}$, with only $a_\pm$ nonzero.
The relations of Eq. (53) generalise to

$$\lambda_\chi = \frac{3}{8}\sum_{\tilde\chi}\beta_{\chi\tilde\chi}\,M_0^{\tilde\chi}\,, \quad a_\chi = \frac{1}{2}\sum_{\tilde\chi}\beta_{\chi\tilde\chi}\,M_1^{\tilde\chi}\,, \quad b_\chi = \frac{1}{8}\sum_{\tilde\chi}\beta_{\chi\tilde\chi}\,M_2^{\tilde\chi}\,, \tag{15}$$

where $\beta_{\chi\chi} = \beta^{\rm intra}_{2,2}$, $\beta_{\chi,-\chi} = \beta^{\rm inter}_{2,2}$, and $M_n^{\tilde\chi}$ is defined as in the single-node case for the node $\tilde\chi$. The coefficients $a_+$ and $a_-$ are coupled through $\beta^{\rm inter}_{2,2}$, which thus acts as an ordinary additional scattering channel for the vertex correction.
The vanishing of $\{\lambda_\chi, b_\chi\}$ also follows from the continuity equation of each node. Multiplying Eq. (43) by $\mathcal{W}_\chi(\gamma)$ and integrating over $\gamma$ gives the rate equation for the number of electrons at the node $\chi$. The driving term contributes the pumping rate $\int d\gamma\,\mathcal{W}_\chi\,h_{\chi,2}$, which vanishes by parity. The intranode part of the collision integral drops out, because the scattering is elastic and $\mathcal{T}_{2,2}(\gamma,\gamma')$ is symmetric under $\gamma \leftrightarrow \gamma'$. The internode part gives the rate of transfer to the other node, which is proportional to $\beta^{\rm inter}_{2,2}\int d\gamma\,d\gamma'\,\mathcal{W}_\chi(\gamma)\,\mathcal{W}_{-\chi}(\gamma')\,\mathcal{T}_{2,2}(\gamma,\gamma')\left[\Lambda^z_{\chi,2}(\gamma) - \Lambda^z_{-\chi,2}(\gamma')\right]$. Since $\mathcal{W}$ is even and the term $4\cos\gamma\cos\gamma'$ of $\mathcal{T}_{2,2}$ averages to zero against it, the odd part of $\Lambda^z_{\chi,2}$ does not contribute to this rate. Only the even part $\tau_{\chi,2}\left[\lambda_\chi + b_\chi\cos(2\gamma)\right]$ can transfer electrons between the nodes. Since the pumping rate vanishes, the homogeneous equations of the even sector have only the trivial solution once the number constraint is imposed. The continuity equation of each node is thus satisfied by

itself, with neither pumping nor transfer. Consequently, $\delta\mu_\chi = 0$, the number constraint holds node by node and not only for the sum, and no charge is pumped between the nodes. The net $\delta\sigma_{zz}$ still depends on the ratio $\beta_{2,2}^{\rm inter}/\beta_{2,2}^{\rm intra}$, as seen in Fig. 5."
An analogous parity argument has been added to the single-node HSM subsection of Sec. III. For a single HSM node, the even sector of the ansatz vanishes by parity, whereas the coefficient of the odd harmonic is nonzero and acts as a vertex correction that is absent in the relaxation-time approximation.
The general arguments on the singularity of the collision matrix $\mathcal{A}$ have been collected in the new Appendix C, and the derivation of the node-resolved continuity equation is given in the new Appendix D. Appendix C contains the zero mode of the collision operator, its uniqueness, the resulting ranks of $\mathcal{A}$, and an explicit check at the level of the matrix elements that the coefficients of the inverse relaxation time form a null vector. For the VNR, the mirror symmetry $k_z \to -k_z$ makes the source of Eq. (49) vanish, so that $\lambda = a = 0$ and the vertex correction is absent for collinear fields.
A short comparison with the BC-monopole nodes of Refs. [1, 2] has been added at the end of the two-node subsection.
The abstract, the Introduction and the Summary have been updated to reflect these points. In the Introduction, we now also refer to the relaxation-time analysis of the VNR and the HSM in Ref. [**?** ].

## III. LIST OF CHANGES

We have marked all additions in the manuscript with the colour red. These comprise a paragraph on the time-reversal symmetry of the bare Hamiltonian versus the parameter-reversal relation in Sec. II, a discussion of the limit $\mu \to 0$ in Sec. III, a subsection with explicit results for the combined in-plane and axial tilt in Sec. IV, together with the new Appendix B containing the intermediate steps, and the revised captions of Figs. 1 to 3 together with a connecting paragraph in the Introduction. The abstract, the Introduction and the Summary have been updated accordingly, including the cancellation of the odd-in-$B_z$ terms in the two-node HSM and the explicit results for the combined tilt. The two-node corrections are described in the preceding section. The new Appendices C and D contain the general discussion of the singularity of the collision matrix and the node-resolved continuity equation, respectively.

[1] A. Knoll, C. Timm, and T. Meng, Negative longitudinal magnetoconductance at weak fields in Weyl semimetals, Phys. Rev. B **101**, 201402 (2020).
[2] I. Mandal, Longitudinal magnetoconductivity in multifold semimetals exemplified by pseudospin-1 nodal points, Eur. Phys. J. Plus **141**, 75 (2026).